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Lecture 1. (YV) On the perversity of affine Springer sheaves and their coinvariants

I. Let G conn. red. gp $/k=\bar{k}$, $(\ell, \mathrm{char}\, k)=1$.

$$D\left(\frac{G}{G}\right) = D^G(G) = \mathrm{Ind}\, D^b_c\left(\frac{G}{G}\right)$$

Let $B \subset G$ Borel, $T = B/R_u(B)$. $W = W_G$ Weyl gp

$$\frac{G}{\mathrm{Ad}\,G} \xleftarrow{\pi} \frac{B}{\mathrm{Ad}\,B} \xrightarrow{r} \frac{T}{\mathrm{Ad}\,T}$$

π proper, r is smooth of rel dim 0

$\forall\, \mathcal{L} \in D\left(\frac{T}{T}\right)$ local system. $\rightsquigarrow$ $S_{\mathcal{L}} = \pi_! r^! (\mathcal{L})$

Fact. (Springer, Lusztig, ...)

1) $S_{\mathcal{L}}$ is perverse

2) if $\mathcal{L}$ is W-equivariant, then $S_{\mathcal{L}}$ is W-equivariant.

3) $\forall\, \tau \in \mathrm{Rep}_{\bar{\mathbb{Q}}_\ell}(W)$, $S_{\mathcal{L},\tau} = \tau \underset{\mathbb{Q}_\ell[W]}{\otimes} S_{\mathcal{L}}$ is perverse.

Reason to extend to LG.

$$LG : \underset{\substack{\| \\ k\text{-}\mathrm{Alg}}}{\mathrm{Aff}_k^{op}} \longrightarrow \mathrm{Sets}, \quad LG(R) = G(R((t)))$$

Let $I \subset LG$ Iwahori, $\tilde{T} = I/I^+$, $\frac{LG}{LG} \xleftarrow{\pi} \frac{I}{I} \xrightarrow{r} \frac{\tilde{T}}{\tilde{T}}$

$\forall\, \mathcal{L} \in D\left(\frac{\tilde{T}}{\tilde{T}}\right)$ loc sys. $S_{\mathcal{L}} = \pi_! r^! (\mathcal{L})$ affine Springer sheaf (π_*)

Let $(LG)_{bd} \subset LG$ locus of "bounded elements"

$$\pi : \frac{I}{I} \twoheadrightarrow \frac{(LG)_{bd}}{LG}$$

If $\mathcal{L}$ is W-equivariant, then $S_{\mathcal{L}}$ is $\widetilde{W} = W \ltimes X_*(T)$ - equivariant (Lusztig)

$\Rightarrow \forall \mathcal{L}, \ \bigoplus_{w \in W} w^*(\mathcal{L})$ is W-equivariant $\Rightarrow \bigoplus_{w \in W} S_{w^*(\mathcal{L})} \circlearrowleft \widetilde{W}$

$\Rightarrow S_{\mathcal{L}} \circlearrowleft X_*(T)$

Th. 1) $\exists$ nat'l t-str. on $D\left(\frac{(LG)_{bd}}{LG}\right)$

2) $S_{\mathcal{L}}$ is perverse

3) $\forall \tau \in Rep(X_*(T))$, $S_{\mathcal{L},\tau} = \tau \overset{L}{\underset{\overline{\mathbb{Q}}_\ell[X_*(T)]}{\otimes}} S_{\mathcal{L}}$ is perverse

if $F = k((t))$.

Rem. need char $k \gg 0$.

II. Prestack. functor $Aff_k^{op} \longrightarrow$ Groupoids

$\mathcal{Y} \mapsto D(\mathcal{Y})$ - Stable ∞-cat. of ind-constr. sheaves.

① $\mathcal{Y} = Y \in Aff_k^{f.t.}$, then $D(Y) = \mathrm{Ind}\, D_c^b(Y)$.

② Let $Y \in Aff_k$, then $Y = \varprojlim Y_\alpha$, $Y_\alpha \in Aff_k^{ft}$, $D(Y) = \mathrm{colim}^{!-pb} D(Y_\alpha)$

③ $\forall \mathcal{Y}$, $D(\mathcal{Y}) = \lim_{\substack{Y \to \mathcal{Y} \\ Y \in Aff_k}}^{!-pb} D(Y)$.

Property: $\forall f: \mathcal{X} \to \mathcal{Y}$, $\exists f^!: D(\mathcal{Y}) \to D(\mathcal{X})$

but other functors $f_!$, f_*, f^* exist only for some f.

Ex. $\pi: \frac{I}{I} \to \frac{LG}{LG}$ ind-proper, $\Rightarrow$ $\exists$ left adjoint $\pi_!$ of $\pi^!$

$\to$ $S_L = \pi_! r^!(\mathcal{I})$ defined.

(III) Perverse t-structures.

Want $\left(D^{\leq 0}(\mathcal{Y}), D^{\geq 0}(\mathcal{Y})\right)$ subcats of $D(\mathcal{Y})$ satisfying some properties.

① Let $Y \in \mathrm{Aff}_k^{ft}$ $\rightsquigarrow$ $\left(D^{\leq 0}(Y), D^{\geq 0}(Y)\right) = \left({}^{Pcl}D^{\leq 0}(Y), {}^{Pcl}D^{\geq 0}(Y)\right)[\dim Y]$

$= \left({}^{Pcl}D^{\leq -\dim(Y)}(Y), {}^{Pcl}D^{\geq -\dim(Y)}(Y)\right)$

Property. if $f: X \to Y$ smooth $\Rightarrow$ $f^!$ is t-exact.

② $Y \in \mathrm{Aff}_k$ is called placid if $Y = \lim Y_\alpha$, $Y_\alpha \in \mathrm{Aff}_k^{ft}$ and $Y_\alpha \to Y_\beta$ are smooth

Ex. If $X \in {}^{\mathrm{Aff}}\mathrm{Sch}^{sm, ft}(k)$, L^+X is placid.

Then $\exists!$ t-str. $D^{\leq 0}(Y) = \operatorname{colim} D^{\leq 0}(Y_\alpha)$

(Y - placid) $\quad$ —"— ≥ 0 $\quad$ —"— ≥ 0, —"—

③ Let $\mathcal{Y} = X/G$, X placid (affine) scheme, G pro-smooth gp scheme

Then $\exists!$ t-structure on $D(\mathcal{Y})$ s.t. $\pi^!: D(\mathcal{Y}) \to D(X)$ is t-exact.

$\overset{\shortparallel}{\lim}\left(D(X) \rightrightarrows D(G\times X) \overrightarrow{\rightrightarrows} D(G\times G\times X) \ldots\right)$

④ $\mathcal{Y}$ is called placidly stratified ∞-stack if $Y = \coprod Y_\alpha$, Y_α placid.

+ $\eta_\alpha : Y_\alpha \to Y$ is f.p. locally closed emb.

s.t. $\exists\ \eta_\alpha^!, \eta_\alpha^*, \eta_{\alpha!}, \eta_{\alpha,*}$ satisfying usual identities.

⑤ Assume $\exists$ open emb. $j: Y^\circ \hookrightarrow Y$ s.t. $\exists\ j_! : D(Y^\circ) \to D(Y)$ f.f.

$\forall$ t-str. on $D(Y^\circ)$, $\exists !$ t-str. on $D(Y)$ s.t. $\mathcal{F} \in D^{\leq 0}(Y) \Leftrightarrow j^! \mathcal{F} \in D^{\leq 0}(Y_0)$

Then $\mathcal{F} \in D^{\geq 0}(Y) \Rightarrow j^! \mathcal{F} \in D^{\geq 0}(Y)$

$\Leftarrow$ holds if $\mathcal{F} \xrightarrow{\sim} j_* j^! \mathcal{F}$.

IV. Let $C = G /\!/ G$ - Chevalley space. $k[C] = k[G]^G \hookrightarrow k[G]$

Let $\chi : G \to C$

Ex. $G = SL_n$, $C = \mathbb{A}^{n-1}$

$$\begin{tikzcd} LG \arrow[r, "\chi"] & LC \\ (LG)_{bd} \arrow[u] \arrow[r] \arrow[ur, phantom, "\urcorner"] & L^+C \arrow[u] \end{tikzcd}$$

Let $D : C \to \mathbb{A}^1$ discriminant, $D(t) = \prod_{\alpha \in \Phi(G,T)} (1-\alpha)$

$$\begin{tikzcd} L^+C \arrow[r, "D"] & L^+\mathbb{A}^1 \\ (L^+C)_\circ \arrow[u, phantom, "\cup"] \arrow[r] \arrow[ur, phantom, "\urcorner"] & L^+\mathbb{A}^1 - \{0\} = L^+(\mathbb{A}^1)_\circ \arrow[u, phantom, "\cup"] \end{tikzcd}$$

$$\begin{tikzcd} (LG)_{bd} \arrow[r] & L^+C \\ (LG)_{bd,\circ} \arrow[u, phantom, "\cup"] \arrow[r] & (L^+C)_\circ \arrow[u, phantom, "\cup"] \end{tikzcd}$$

$(LG)_{bd,\circ} = LG^{rss}_{bd}$

Claim $\frac{(LG)_{bd,\bullet}}{LG}$ - placidly stratified

Pf. $(L^+\mathbb{A}^1)_\bullet = \coprod (L^+\mathbb{A}^1)_n$ ← val. $= n$

$\Rightarrow (L^+C)_\bullet = \coprod (L^+C)_\alpha$ $\quad (L^+C)_\alpha$ = conn'd component of $\mathbb{D}^{-1}((L^+\mathbb{A}^1)_n)$

↑

Goresky - Kottwitz - MacPherson

$\Rightarrow (LG)_{bd,\bullet} = \coprod (LG)_{bd,\alpha}$

Prop. $\frac{(LG)_{bd,\alpha}}{LG}$ is a placid stack $\frac{(L^+C)_\alpha}{L^+T_\alpha \times \mathbb{Z}^r}$

If $(Y, (Y_\alpha)_\alpha)$ - placidly stratified, $\rightarrow$ $\exists$ t-str. on $D(Y)$

$\mathcal{F} \in D^{\le 0}(Y) \iff \eta_\alpha^* \mathcal{F} \in D^{\le -\nu(\alpha)}(Y_\alpha)$

$\eta_\alpha^! \mathcal{F} \in D^{\ge -\nu(\alpha)}(Y_\alpha)$ $\quad \forall \alpha$

for every $\nu : A \to \mathbb{Z}$

nat'l perversity $\Rightarrow \nu(\alpha) = \operatorname{codim}_Y(Y_\alpha)$

$$\begin{tikzcd} \frac{I}{I} \arrow[d, "\pi"', "\text{proper birat'l}"] & \frac{I_\alpha}{I} \arrow[l] \arrow[d, "\pi_\alpha", "\text{equidim morphism with fin dim fibres}"'] \\ \frac{(LG)_{bd}}{LG} & \supset \frac{(LG)_{bd,\alpha}}{LG} \end{tikzcd}$$

$\operatorname{codim}_{\frac{(LG)_{bd}}{LG}}\left(\frac{(LG)_{bd,\alpha}}{LG}\right) := \dim \pi_\alpha + \operatorname{codim}_I(I_\alpha)$

Th. S_L is preverse $\Leftarrow$ π is semismall

$\iff \dim \pi_\alpha \le \operatorname{codim}_I(I_\alpha), \quad \forall \alpha \quad . \quad I \to L^+C$ flat

Lecture 2 (YV)

Motivation: f.d. case

Let G conn. reductive gp $/\mathbb{F}_q$, $(\ell, q)=1$,

$V = \overline{\mathbb{Q}}_\ell[G(\mathbb{F}_q)]^{G(\mathbb{F}_q)}$ has a nat'l basis χ_ρ, $\rho \in \mathrm{Rep}^{irr}_{\overline{\mathbb{Q}}_\ell}(G(\mathbb{F}_q))$.

Lusztig. 1) V has another nice basis "coming from geometry", that is

$\{\chi_A^a\}$, A is Frob equiv. char. sheaf on G.

2) $\exists$ explicit formulae $(\chi_\rho) \rightsquigarrow (\chi_A^a)$.

Summary. $\exists$ explicit lin. comb. of χ_ρ "coming from geometry".

Ex 1. $G = GL_n$, $\rho \in \mathrm{Irr}(GL_n(\mathbb{F}_q))$ s.t. $\rho^{B(\mathbb{F}_q)} \neq 0$

Then $\rho \rightsquigarrow \rho^{B(\mathbb{F}_q)} \in \mathrm{Irr}\,\mathcal{H}(G(\mathbb{F}_q), B(\mathbb{F}_q)) \rightsquigarrow \tau_\rho \in \mathrm{Irr}(W)$

$$\mathcal{H}(G(\mathbb{F}_q), B(\mathbb{F}_q)) = H_q \simeq \overline{\mathbb{Q}}_\ell[W]$$

Th. (Lusztig) $\chi_\rho = \mathrm{Tr}(Fr, S_{\tau_\rho})$ if $G = GL_n$. $S = S_{\overline{\mathbb{Q}}_\ell}$ - Grothendieck Springer sheaf

$$S_{\tau_\rho} = \tau_\rho \underset{\overline{\mathbb{Q}}_\ell[W]}{\otimes} S$$

Ex 2. Let $T \subset G$ max torus, $\theta : T(\mathbb{F}_q) \to \overline{\mathbb{Q}}_\ell^\times$

$\rightsquigarrow DL(T,\theta) \in D^b_c(\mathrm{Rep}\, G(\mathbb{F}_q))$. $\theta \rightsquigarrow \mathcal{L}_\theta$ loc. sys. on T

Choose "admissible" isom $T \xrightarrow[\sim]{\varphi} T_G = B/R_u(B)$, $\mathcal{L}_{\theta,\varphi} := \varphi_* \mathcal{L}_\theta \in D(T_G)$

$Fr^* \mathcal{L}_{\theta,\psi} \simeq w^* \mathcal{L}_{\theta,\psi} \Rightarrow S_{\mathcal{L}_{\theta,\psi}}$ is Frob-equivariant.

Th. (Lusztig) $\chi_{DL(T,\theta)} = \mathrm{Tr}(Fr, S_{\mathcal{L}_{\theta,\psi}})$.

II. Let F na local field

$\forall \pi \in \mathrm{Rep}^{irr}_{sm}(G(F)) \rightsquigarrow \chi_\pi \in \hat{C}(G(F))^{G(F)}$

inv. generalized function

Q. Does $\mathrm{Span}_{\bar{\mathbb{Q}}_\ell}(\chi_\pi)$ has a basis coming from geometry?

A1: Don't know (even conjecturally)

A2: Expect it is true for $\mathrm{Span}(\chi_\pi|_{G(F)_c})$

Rem For unip. rep, $\exists$ precise conj. by Lusztig.

Langlands conj.: 1) $\exists$ decomposition $\mathrm{Rep}^{irr}_{sm}(G(F)) = \coprod_\lambda \Pi_\lambda$ - L-packets

2) $\forall \lambda$, $\mathrm{Span}(\chi_\pi)_{\pi \in \Pi_\lambda}$ has another basis

χ_λ^k s.t. χ_λ^k is "$\mathcal{E}_{\lambda,k}$- stable", i.e. χ_λ^k has nice arithmetic properties.

Expectation:

1) $\forall$ Frob-equiv. affine char sheaf $\mathcal{A} \in D\left(\frac{LG_{comp}}{LG}\right)$, $\chi'_\mathcal{A} := \mathrm{Tr}(Fr, \mathcal{A})$

is $\mathcal{E}_\mathcal{A}$- stable.

2) $\chi_\lambda^k|_{G(F)_{comp}}$ = linear comb. of $\mathcal{E}$- stable.

3) $\chi_\lambda^k|_{G(F)_{comp}}$ is $\mathcal{E}$- stable $\Rightarrow \chi_\lambda^k$ is $\mathcal{E}$- stable (known by Arthur.. if char $F = 0$)

Main results G - split

1) Let $\pi \in \mathrm{Rep}^{irr}_{sm}(G(F))$ s.t. $\pi^I \neq \{0\}$

Then $\pi \rightsquigarrow \pi^I \in \mathrm{Irr}\, H(G(F), I) = \mathrm{Irr}\, H_q$

$$H_q \xrightarrow[q \to 1]{} \overline{\mathbb{Q}}_\ell[\tilde{W}]$$

$$\pi^I \underset{q \to 1}{\rightsquigarrow} \tau_\pi \in \mathrm{Rep}_{\overline{\mathbb{Q}}_\ell}[\tilde{W}]$$

Let $S_\tau = \tau \overset{L}{\underset{\overline{\mathbb{Q}}_\ell[\tilde{W}]}{\otimes}} S$ - derived coinvariants

Th (Bezrukavnikov - Kazhdan - V.) (version of Lusztig conjecture)

If $G = GL_n$, $\forall \pi \in \mathrm{Rep}^{irr}_{sm}(G(F))$, $\pi^I \neq 0$,

$$\chi_\pi|_{G(F)_c} = \mathrm{Tr}(Fr, S_\tau). \in \hat{C}(G(F)_c)^{G(F)}$$

In particular, $\forall \gamma \in G(F)^{rss}_c$

$$\chi_\pi(\gamma) = \mathrm{Tr}\Big(Fr, \tau_\rho \overset{L}{\underset{\overline{\mathbb{Q}}_\ell[\tilde{W}]}{\otimes}} R\Gamma_c(Fl_\gamma, W_{Fl_\gamma})\Big)$$

$$= \sum_{i,j} (-1)^{i+j}\, \mathrm{Tr}\big(Fr, \mathrm{Tor}_i^{\tilde{W}}(\tau_\rho, H^j_c(Fl_\gamma, W_{Fl_\gamma}))\big)$$

Rem Th Yun $\Rightarrow H^j_c(Fl_\gamma, W_{Fl_\gamma})$ is f.g. $\tilde{W}$ - module

$\Rightarrow \mathrm{Tor}_i(\tau_\rho, H^j_c(\ldots))$ is f.d. vector space $\Rightarrow$ $\mathrm{Tr}(Fr, -)$ is defined

Cuspidal DL rep'n. $Z(G)(F)$ - compact

G - split $/ F^{nr}$, $T \subset G$ elliptic, $\Rightarrow T(F) = T(\mathcal{O})$
unram. max'l torus

$\theta : T(F) = T(\mathcal{O}) \longrightarrow \overline{T}(\mathbb{F}_q) \xrightarrow{\bar{\theta}} \overline{\mathbb{Q}}_\ell^\times$. Assume $\bar{\theta}$ is in general position.

Pages

$\exists!$ parahoric subgp $T(\mathcal{O}) \subset G_{x,0}$

$$T \hookrightarrow G \rightsquigarrow T(\mathcal{O}) \hookrightarrow G_x \rightsquigarrow \bar{T} \subset L_x \quad , L_x(k) := G_{x,0}/G_{x,0^+}$$

$$\pm\, DL(\bar{T}, \bar{\theta}) \in \mathrm{Rep}^{cusp}_{irr} L_x$$

$$DL(\bar{T}, \bar{\theta}) \rightsquigarrow \underset{\substack{\uparrow \\ \mathrm{Rep}^{cusp}_{irr} G(F)}}{\pi_{T,\theta}} := \mathrm{ind}_{G_x}^{G(F)} \mathrm{inf}_{L_x(\mathbb{F}_q)}^{G_x} DL(\bar{T}, \bar{\theta})$$

Let $a_1, \dots, a_n : T \hookrightarrow G$ set of rep'n of $T \hookrightarrow G$ s.t. conj. to $a_0 : T \hookrightarrow G$ inclusion.

$$\forall a_i \rightsquigarrow \pi_{a_i(T),\theta} \in \mathrm{Rep}^{irr}_{cusp}(G(F)).$$

$$\forall \kappa \in \hat{T}^{\underset{\substack{\| \\ \Gamma_F}}{Gal(\bar{F}|F)}} \rightsquigarrow \chi^{\kappa}_{T,\theta} := \sum_i \langle \kappa, \underset{\substack{\uparrow \\ H^1(F,T) = X_*(T)_{\Gamma_F}}}{inv(a_i : a_0)} \rangle \chi_{\pi_{a_i(T),\theta}}$$

Th. $\chi^{\kappa}_{T,\theta} = \mathrm{Tr}(Fr, S_{\mathcal{L}_{\theta,\varphi}, \kappa})$

$$\mathcal{L}_{\theta,\varphi} \in D\left(\frac{\hat{T}}{T}\right) \rightsquigarrow S_{\mathcal{L}_{\theta,\varphi}} - \text{Affine Groth. Spr.}$$

$$\frac{\bar{T}}{\bar{T}} \leftarrow \frac{I}{I} \longrightarrow \frac{LG}{LG}$$

Cor: $\chi^{\kappa}_{T,\theta}$ is Σ_k - stable.

Lecture 3 (RB) $\quad G$ split reductive$/F$, $F = \mathbb{F}_q((t))$, T, θ - local system

$$\exists!\, w, \quad Fr^*(\theta) = w(\theta) \quad , w\text{ - elliptic}.$$

R_θ - cuspidal DL rep'n of $G(\mathbb{F}_q)$

Cuspidal L- packet $= \{ c\text{-}\mathrm{Ind}_{P_i}^{G}(R_\theta) \}$, $P_i = G_{x_i}$, $\underset{\text{conj. class}}{(w)} \cap W_{x_i}$

$\Lambda = X_*(T)$, $\qquad \{x_i\} \longleftrightarrow \left(t_{\mathbb{R}}/\Lambda\right)^{w} \simeq \Lambda_w$

OTOH, consider the CS on G (f.dim.), $\mathcal{F}_\theta$,

parahoric induction to LG gives a sheaf $\widetilde{\mathcal{F}}_\theta \supseteq \Lambda$.

For $x: \Lambda \to \mathbb{C}^\times$, $\qquad \widetilde{\mathcal{F}}_{\theta, x} = \widetilde{\mathcal{F}}_\theta \underset{\mathbb{C}[\Lambda]}{\otimes} x$ - CS on LG

$x \in T^\vee(\mathbb{C})$

If $x \in (T^\vee)^w$, $\mathcal{F}_{\theta, x}$ carries a Fr-structure.

Ex. $\underset{\{x\}}{\overset{\|}{(T^\vee)^w}} = (\Lambda_w)^*$

Th. a) characters of L-packet members $\xleftrightarrow{FT}$ $\operatorname{Tr} \operatorname{Fr}(\overset{\widetilde{\mathcal{F}}_{\theta, x}}{\overset{\|}{\mathcal{I}_{\theta, x}}})$

b) $\operatorname{Tr} \operatorname{Fr}(\widetilde{\mathcal{F}}_{\theta, x})$ are endoscopic

$x = 1$ - Stable $\qquad \Leftarrow$ Yun's Thm.

$\forall \gamma \in LG^{rs}$, $(\widetilde{\mathcal{F}}_{\theta, x})_\gamma \supseteq \underset{\text{Centralizer}}{Z(\gamma)}$ acts trivially

Unipotent characters / CS (joint w/ Ciubotaru, Kazhdan, Varshavsky)

G - p-adic group, $\qquad H_I = C_c(I \backslash G / I)$

$$\underset{\text{cocentre}}{C(G)} = \underset{\overset{\wr\wr}{\ddot{H}}}{C_c^\infty(G)_G} = H/[H,H] \ominus \underset{\text{unipotent}}{C_{unip}} \ominus H_I/[\,,\,] \;\; = G \text{ of type } A.$$

$\mathcal{H} = Sh(I \backslash LG / I)$

unip. monodromic

$$\mathcal{C}(\mathcal{H}) = hh(\mathcal{H}) \xrightarrow[\sim]{BZ, N, P} D\,Coh_{\mathcal{N}}^{\check{G}}(Z)$$

G simply conn'd

$Z \subset \check{\mathfrak{g}} \times \check{G}$ commuting (derived) scheme

Supp. on $\{(e, g) : e \in \mathcal{N}\}$ nilp. cone.

Th. $C_{unip} \simeq K_q(\mathcal{C}(\mathcal{H}))$

$$K_q = K\, Coh_{\mathcal{N}}^{\check{G} \times \mathbb{G}_m}(Z) \underset{\mathbb{Z}[t, t^{-1}]}{\otimes} \mathbb{C} \qquad t \mapsto q$$

$\mathbb{G}_m$ - dilates e $\qquad \mathbb{Z}[t, t^{-1}] = K(\mathbb{G}_m\text{-mod})$

Can rewrite it using only the action of $q^{\mathbb{Z}} \subset \mathbb{G}_m$

$$= K\, Coh_{\mathcal{N}}^{\check{G} \times q^{\mathbb{Z}}}(Z) \underset{K(\mathbb{Z}\text{-mod})}{\otimes} \mathbb{C}$$

<u>Idea of proof</u> $\quad J$, $\quad P_J$ - standard parahoric

$$H > H_P = C\left(\frac{P_J^+ \backslash G / P_J^+}{L_J}\right)_{unip}$$

Recall $hh(\mathcal{H}_J, \mathcal{H})$ - parahoric $CS =: \mathcal{H}_P$

Results of Lusztig $\Rightarrow K_q(\mathcal{H}_P) \simeq H_P$. (using $\mathbb{Z}$-action by Fr)

Lemma (exercise on Schneider - Stuhler)

$$C_{unip} = \operatorname*{colim}_{\substack{J \\ J^c \neq \phi}} H_{P_J} = \operatorname*{colim}_{\substack{J \\ J^c \neq \phi}} H_{P_J} / [\,,\,]$$

From this get a map LHS $\to$ RHS.

$$\begin{tikzcd}
\mathcal{H} \arrow[rr] \arrow[dr] & & \mathcal{C}(\mathcal{H}) = hh(\mathcal{H}) \\
& \mathcal{H}_{P_J} = hh(\mathcal{H}_J, \mathcal{H}) \arrow[ur] &
\end{tikzcd}$$

get $K_q(\mathcal{H}_{P_J}) \to$ RHS. $\qquad$ get $\operatorname{colim} H_{P_J} \to$ RHS

$\quad\; \| \qquad\qquad\qquad\qquad\qquad\qquad\qquad \perp \; \|$

$\quad H_{P_J} \qquad\qquad\qquad\qquad\qquad\qquad\quad C_{unip}$

need to see it's injective, onto

<u>inj</u>: suppose $c \in C_{unip}$, $c \mapsto 0$. want $c = 0$.

Enough to see $O_\gamma(c) = 0$, $\forall \gamma \in G^{rs}$

O_γ comes from a commutator functor $\mathcal{H} \to Vect$

conclude $O_\gamma(c) = 0$.

<u>surj</u>. proof 1, reduce to fin. dim subspaces, compare dim.

proof 2. use comm. diagram

$G \supset P$ parahoric, $\bar{P}$ – fin. dim'l reductive gp

$$\begin{tikzcd}
\mathcal{F} \in CS(\bar{P}) \arrow[r, hook] & \mathcal{H}_P \arrow[d] \\
& \mathcal{C}(\mathcal{H}) = D\,Coh^{\check{\hat{h}}}_{\mathcal{N}}(Z)
\end{tikzcd}$$

unipotent CS

$\bar{P} = Z_G(s)$, $s \in G$ $\qquad$ Fix $\theta_s^\vee$ – local system on $\check{T}$ with the same stabilizer.

Lusztig - Yun $\Rightarrow$ $CS_{unip}(\bar{P}) \simeq CS_\theta(G^\vee)$

$$\mathcal{F} \longmapsto \mathcal{F}^\vee \text{ - Hodge module}$$

Prop. (w Losev - Ma), $\mathcal{F} \mapsto gr_{Hodge}(\mathcal{F}^\vee)$

"compute" RHS

Th 2. $K_q(Coh^{G^\vee}_N(Z)) = \bigoplus_{e \in N/\sim} K\,Coh^{Z_e}(Z_e)$

$$Z_e = Z(e)_{red}$$

$Z(e)$ - centralizer

$$\{e\} \times Z_e \xhookrightarrow{i_e} Z$$

$$[\mathcal{F}] \longmapsto ([i_e^* \mathcal{F}]) \in K_q^{Z_e}(Z_e) \simeq K^{Z_e}(Z_e)$$

(only works with K_q, q not root of 1)

Prop H - red. ($H = Z_e$) $K^H(H) \xrightarrow{\sim} \mathcal{O}_e(Comm(H))$ - regular H-inv. func, loc. const in x.

$Comm(H) = \{(x,y) \in H \times H : xy = yx\}$

$$\mathcal{F} \longmapsto f_{\mathcal{F}}(x,y) = Tr(y, \mathcal{F}_x)$$

Next step: describe characters of standard modules. almost char. (dist. coming from CS on LG)

+ endoscopic properties.

$G \supset G^c$ - compact (bounded) elts

$C(G^c)_G \subset C(G)$ $\quad C^c_{uni} = C^c \cap C_{unip}$

Th $\quad C_{unip} \simeq K_q\left(Coh_{\mathcal{N}}^{G^\vee}(Z)\right) \simeq \bigoplus_e \mathcal{O}_\ell(Comm(Z_e))$

$\cup$

$C^c_{unip} \qquad \simeq \qquad \bigoplus_e \mathcal{O}_{\ell r}(Comm(Z_e))$

loc. const in both x & y

Rem. $\mathcal{O}_{\ell r}$ carries an involution $x \leftrightarrow y$.

Expect it to swap characters$\big|_{C^c_{unip}}$ & almost chars.

Lecture 4 (RB) $\quad G$ split s.c. $/ F \simeq \mathbb{F}_q((t))$

$$C_{unip} \simeq K_q\left(Coh_{\mathcal{N}}^{G^\vee}(Z)\right) \simeq \bigoplus_{e\in\mathcal{N}/\sim} K^{Z_e}(Z_e) \simeq \bigoplus_{e\in\mathcal{N}/\sim} \mathcal{O}_\ell(Comm(Z_e))$$

unipotent cocentre

commuting variety in $\mathfrak{g}^\vee \times G^\vee$

conj. inv't regular function on commuting pairs (x,y), locally constant in x

x - endoscopic

$y \approx$ Frobenius

$$[\mathcal{F}] \mapsto f_{\mathcal{F}}(x,y) = Tr(y, \mathcal{F}_x)$$

Description of characters. Given (s,e,ψ), $s \in G^\vee_{ss}$

can define $\chi_{s,e,\psi}$: RHS $\longrightarrow \mathbb{C}$

$$f = (f_e)_{e\in\mathcal{N}/\sim} \longmapsto \left\langle f_e \big|_{\{(x,y):\, y=s\}}, \psi \right\rangle$$

class function on $\pi_0(Z_{e,s})$.

Conj. $\chi_{s,e,\psi}$ is the character of the standard rep'n $P_{s,e,\psi}$.

Th (add-ly j+ w/ V. Krylov) This is true if $P_{s,e,\psi}$ is gen'd by I-inv'ts.

Rem 1. In that case, can work with Hecke rep'n $[K(Coh(B^s_e)) : \psi]$. Get Th by "diagram chase".

$1\frac{1}{2}$: std rep'ns of H_{aff} come from irr rep'ns of J - asymp. Hecke algebra.

Th (B., Karpov, Krylov) & O. Popp
$$J = \bigoplus_{e \in \mathcal{N}/\sim} K^{Z_e}\left(\mathcal{B}_e^{\mathbb{G}_m} \times \mathcal{B}_e^{\mathbb{G}_m}\right)$$

$\curvearrowright Z$

Approach 1. $\rho = \rho_{s,e,\psi}$ - unipotent, $\rho^{P_+} \neq 0$ for some parahoric P.

$\circlearrowleft$

$D(P_+ \backslash LG / P_+)$ Run into a non-~~restricted~~ finite type situation.

Approach 2. work with $Z(\hat{\mathcal{C}}_p)$, categorifying matrix coeff of discrete series, with infinite, converging convolution.

$$\mathcal{C}_p = D\left(\frac{P_+ \backslash LG / P_+}{L}\right)_{unip}$$

Th. $C^c_{unip} \simeq \bigoplus_{e \in \mathcal{N}/\sim} \mathcal{O}_{\ell r}(\mathrm{Comm}(Z_e))$

$\circlearrowleft$
i

Conj. $i(\chi_{s,e,\psi}|_{C^c_{unip}}) = \mathrm{Tr}(Fr, \mathcal{F}_{s,e,\psi})$

character sheaves on the loop group.

Given s, e, ψ, get a functor $\phi_{s,e,\psi} : \mathcal{C} \longrightarrow Vect$

$\mathcal{C} = DCoh^{\check{G}}_{\check{\mathcal{N}}}(Z)$

$\mathcal{F} \longmapsto \mathrm{Hom}_{Z_e}(\psi, \mathcal{F}_{s,e})$

$$LHS \xleftarrow{\mathrm{Tr}\,Fr} \phi_{s,e,\psi}$$

OTOH, starting from s,e,ψ, can produce an object in $D\left(\frac{LG}{LG}\right)$ in another way. Say, if

$[H_*(\mathcal{B}_e^s) : \psi] \neq 0$, the $s,e,\psi \rightsquigarrow$ rep of $H_{aff} \rightsquigarrow$ rep of W_{aff} $R_{s,e,\psi}$

$\mathcal{S}$ - affine Springer sheaf $\curvearrowleft W_{aff}$, $\mathcal{S}_{s,e,\psi} = \mathcal{S} \overset{L}{\underset{W_{aff}}{\otimes}} R_{s,e,\psi}$

can be generalized to other s, e, ψ.
Pagers

Rem. $\mathcal{S}_{s,e,\psi}$ does not represent $\phi_{s,e,\psi}$.

But conj. $\Leftrightarrow$ they give the same functional on C^c_{unip} (up to scalar)

Rem. To check this, enough to compute pairing with $\mathcal{F} \in CS_{unip}(\bar{P})$, P - parahoric

$\langle \mathcal{S}_{s,e,\psi}, \mathcal{F} \rangle$ - computed using orthogonality for CS on $\bar{P}$.

depends on $res_P(\mathcal{S}_{s,e,\psi})$

parahoric restriction

Compared to the answer from $\phi_{s,e,\psi}$ using description of $\mathcal{F}$ in $Coh^{\check{G}}(Z)$ from last time: $gr(\mathcal{F}^\vee)$

$\mathcal{Z}$

General conj. Local Langlands for cocentre.

Recall, a local Langlands parameter is a triple (σ, u, φ)

$\Gamma = Gal(F^{un})$, $\sigma : \Gamma \to \check{G}$, u-unipotent, φ - Frob

$I = \{(\sigma, u)\}$, $Z/\check{G}$ - inertia stack. $Z = (\sigma, u, \gamma)$ $\gamma \in \check{G}$ commutes w σ, u.

φ - conjugation w Fr

Conj. $C \simeq K_\phi(Z/\check{G})$, $K_\phi = K(Z/\check{G} \times \phi^{\mathbb{Z}}) \otimes_{K(\mathbb{Z}\text{-mod})} \mathbb{C}$

$= \bigoplus_{(\sigma,u)/\sim} \mathcal{O}_\ell(Com^\phi(Z_{\sigma,u}))$

$Z_{\sigma,u} = Stab_{\check{G}}(\sigma, u)$, $J_{\sigma,u} = \{ g \in \check{G} : g : (\sigma,u) \to \phi(\sigma,u) \}$

$\{(\tau, z) : z(\tau) = \tau\} = Com^\phi(Z_{\sigma,u}) \subset T_{\sigma,u} \times Z_{\sigma,u}$

Rem. In particular, the depth 0 generic L-packet should be generalized to higher depth based on CS on parahoric subgps

Ex1 σ triv. u-unip. $\phi : u \mapsto u^q$
$u = \exp(e)$

Using JM identifying $\mathcal{O}_\ell(Com^\phi) = \mathcal{O}_\ell(Com(Z_e))$

Ex2. $\Gamma \to \hat{\mathbb{Z}}_{(p)}(1) \to \check{G}$ given by a rss elt σ, $\sigma^q \sim \sigma$, $\sigma^q = w(\sigma)$ for unique w, $Com^\phi = (\check{T})^w \times w\check{T}$

recover $\mathcal{O}_\ell(Com^\phi) = \mathbb{C}[(\check{T})^w]$ $\check{T}$-inv. func. factor through $(\check{T})^w$

Hope. deduce this from coherent description of (restricted) $hh(Sh(IG))$